NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

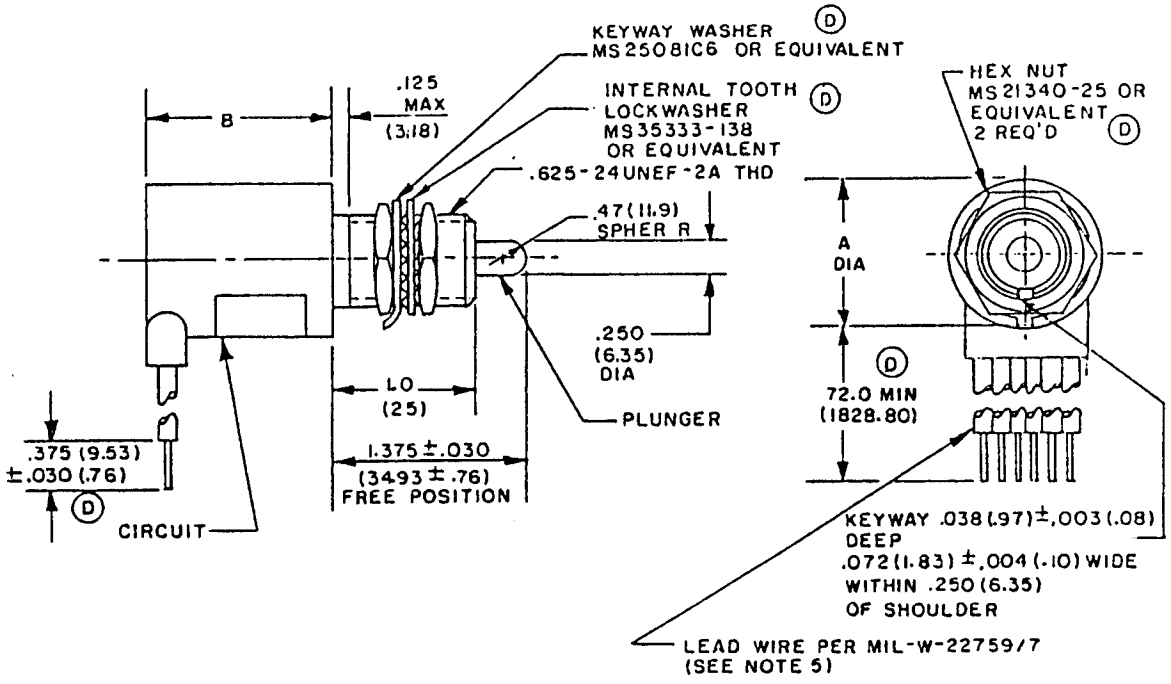
MIL-PRF-8805/80D(USAF)
28 June 1988
SUPERSEDING
MIL-S-8805/80C(USAF)
20 February 1980

PERFORMANCE SPECIFICATION SHEET

SWITCH, SENSITIVE, LIMIT, HERMETIC SEAL

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-8805.



NOTES:

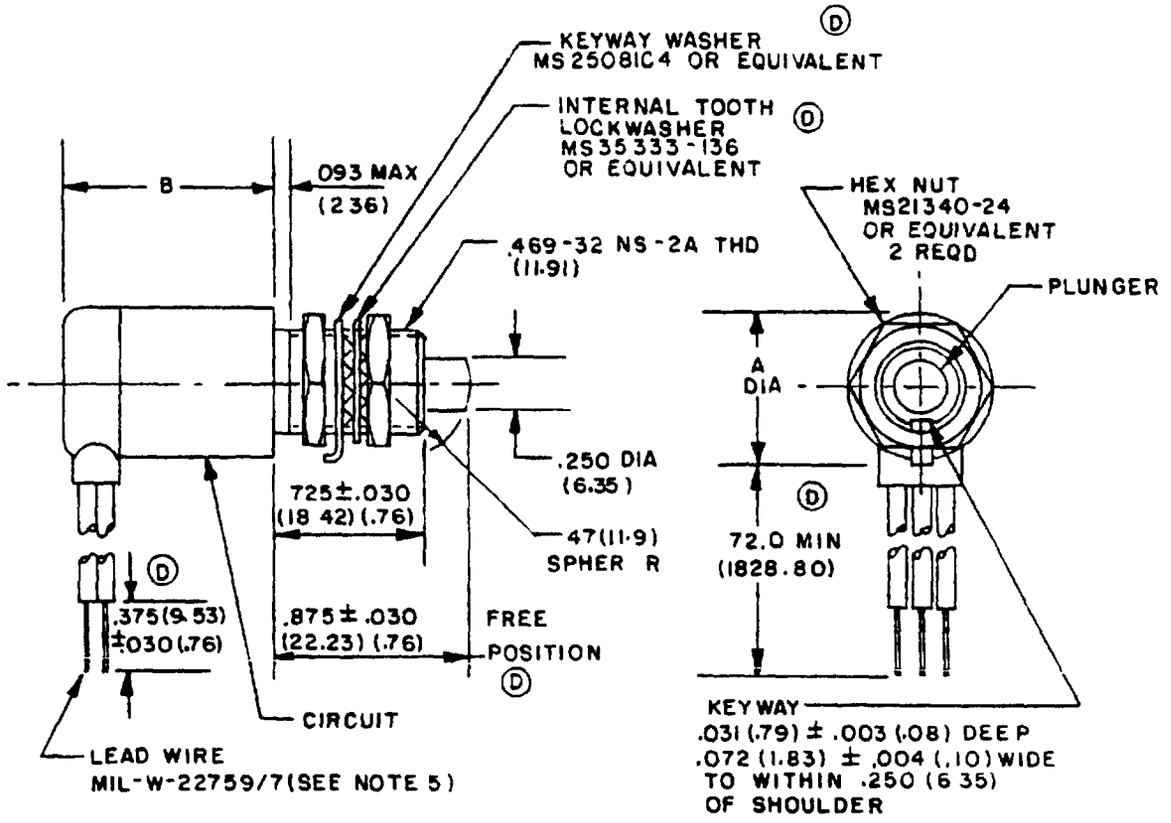
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (.8 mm) for one place decimals, ±.02 (.5 mm) for two place decimals, and ± .005 (.13 mm) for three place decimals.
4. Configuration optional provided specified dimensions are not exceeded.
5. Lead wire shall be marked at intervals of 3.5 inches (89 mm) maximum with switch circuit identification number followed by wire guage number (1-20, etc.). The marking shall be legible, permanent, and shall not impair the wire characteristics.
6. See table III for A and B dimensions.
7. Metric equivalents are in parentheses.

FIGURE 1 . Pin plunger switch with .625 inch (15.88 mm) threaded bushing.

Ⓓ denotes changes

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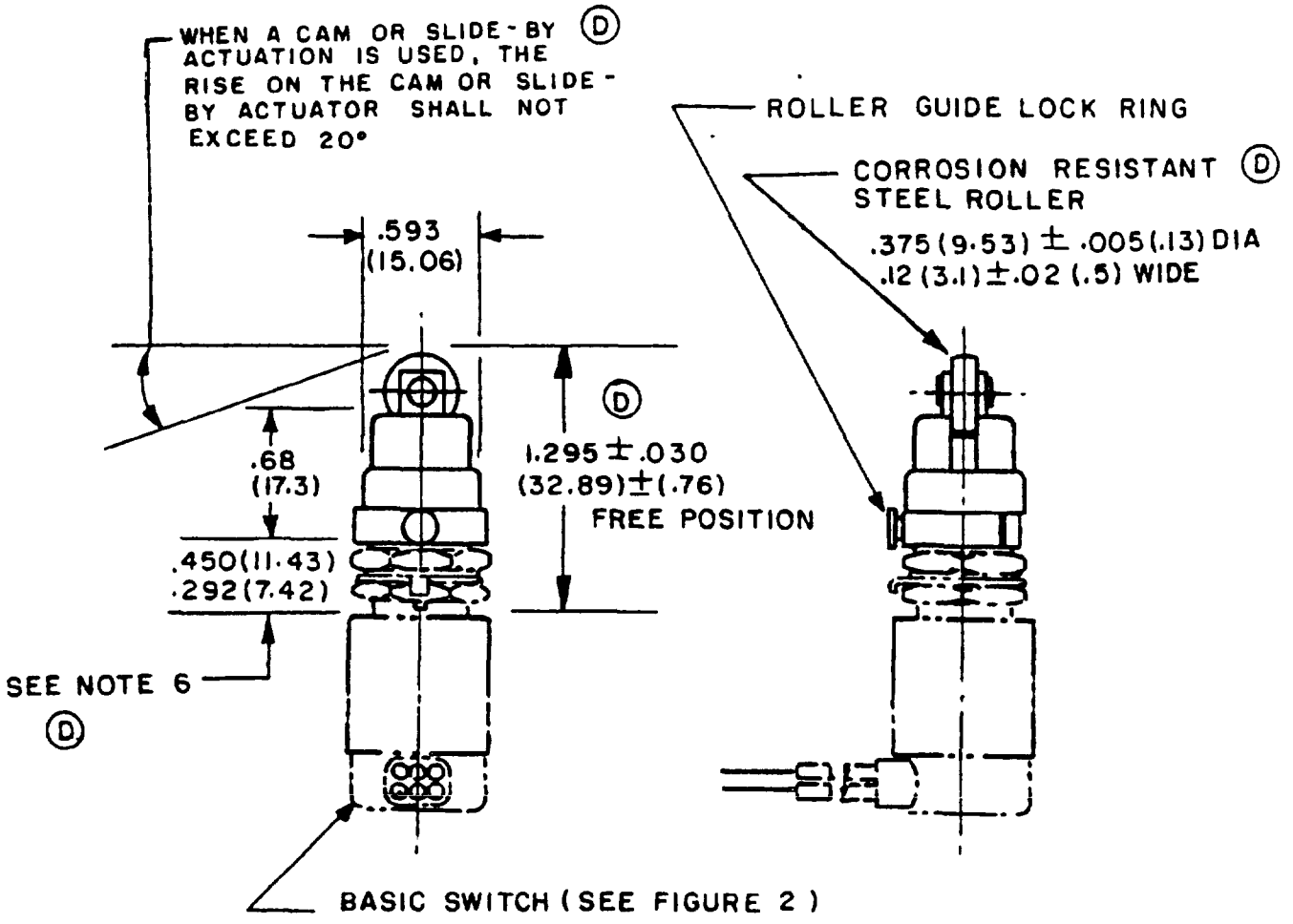
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (.8 mm) for one place decimals, ±.02 (.5 mm) for two place decimals, and ±.005 (.13 mm) for three place decimals
4. Configuration optional provided specified dimensions are not exceeded.
5. Lead wire shall be marked at intervals of 3.5 inches (89 mm) maximum with switch circuit identification number followed by wire gauge number (1-20, etc.). The marking shall be legible, permanent, and shall not impair the wire characteristics.
6. See table III for A and B dimensions.
7. Metric equivalents are in parentheses.

FIGURE 2. Pin plunger switch with .469 inch (11.91 mm) threaded bushing.

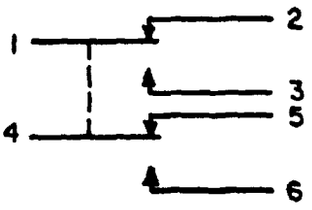
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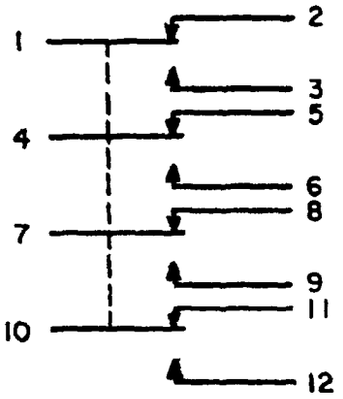
NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (.8 mm) for one place decimals, ±.02 (.5 mm) for two place decimals, and ±.005 (13 mm) for three place decimals
4. Configuration optional provided specified dimensions are not exceeded.
5. Lead wire shall be marked at intervals of 3 5 inches (89 mm) maximum with switch circuit identification number followed by wire gauge number (1-20, etc.). The marking shall be legible, permanent, and shall not impair the wire characteristics.
6 Roller guide shall permit positioning and locking of plunger in increments of 45° azimuth Roller guide location can vary between .292 (7 42 mm) and 450 (11 43 mm) while maintaining .125 (3 18 mm) minimum over travel.
7 Metric equivalents are in parentheses.

FIGURE 3. Roller plunger switch with 469 inch (11 91 mm) threaded bushing.

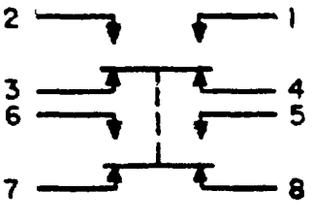
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TYPE A



TYPE B



TYPE C

FIGURE 4. Circuit diagrams.

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REQUIREMENTS:

Dimensions and configuration: See figures 1 through 3.

Enclosure design: 5 (hermetic)-switch chamber.

(D) Temperature characteristics: -55°C to +125°C.

Shock type:

M8805/80-01, and 03 through -08: MIL-STD-202, method 213, half-sine shock pulse of 200 g amplitude with .007 second duration.
M8805/80-02: M (100 g).

Vibration grade:

M8805/80-01, and -03 through -08: 2 (10-2000 Hz except at 20 G's).
M8805/80-02: 1 (10-500 Hz).

Weight with leads:

M8805/80-01, -03, and -07: 7.5 ounces maximum.
M8805/80-02 and -08: 13.5 ounces maximum.
M8805/80-04 and -06: 12.5 ounces maximum.
M8805/80-05: 6.5 ounces maximum.

Operating characteristics:

(D) Operating force:

M8805/80-01 and -02: 9 ±3 pounds.
M8805/80-03 through -08: 8 pounds 8 ounces ±3 pounds 8 ounces.

Full overtravel force: 30 pounds maximum.

(D) Release force:

M8805/80-01 and -02: 4 pounds minimum.
M8805/80-03 through -08: 3 pounds minimum.

Pretravel:

M8805/80-01 and -03 through -08: .040 inch maximum.
M8805/80-02: .070 inch maximum.

Movement differential:

M8805/80-01 and -03 through -08: .020 inch maximum.
M8805/80-02: .045 inch maximum.

Overtravel:

M8805/80-01 through -04: .250 inch minimum.
M8805/80-05 through -08: .125 inch minimum.

Coincidence of operating and releasing points: .010 inch maximum.

} ±20 percent variation from initial values acceptable after test.

Material:

Lockwasher: Stainless steel.

Finish: Switch housing shall be processed to resist corrosion.

Plastic: M8805/80-02 may use general purpose mineral and flock filled phenolic internally.

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Strength of actuating means. 250 pounds.

Strength of mounting bushing:

M8805/80-01 through -04: 400 pound-inches.
M8805/80-05 through -08: 200 pound-inches.

Contact resistance. Not applicable.

Dielectric withstanding voltage:

Sea level: 1,000 V rms. In qualification inspection table after electrical endurance, the dielectric withstanding voltage points of application between all unconnected terminals of the same pole is not applicable.

Fluid resistance: Except for the cut end of the lead wire, switches shall be submerged in each of the following fluids for 2 to 2-1/2 minutes, which shall consist of one cycle (one cycle is 10 to 12-1/2 minutes total). Each switch shall be subjected to three cycles.

a. Turbine fuel (MIL-T-5624).

b. Hydraulic fluid (MIL-H-83306).

c. Coolanel (NSN 9150-00-551-4022).

d. Ethylene glycol (MIL-E-9500).

e. Lubricating oil (MIL-L-7808).

(D) After each immersion, the excess fluid shall be blown off the external surfaces of the switch with an air jet. Following the third cycle, the switch shall be subjected to and shall meet the requirements for dielectric withstanding voltage, insulation resistance, operating characteristics, seal, and marking

Seal test: Only watertight test shall be performed in group A.

Sand and dust test: Applicable.

Icing test:

M8805/80-01: Immersion cycles -3.
M8805/80-02 through -08: Not applicable.

(D) Intermediate current.

M8805/80-01: 100,000 cycles.
M8805/80-02 through -08: Not applicable.

(D) Mechanical endurance: 100,000 cycles.

Electrical endurance: 25,000 cycles.

Electrical ratings: See table I.

Qualification-group submission: See table II.

Military part numbers: See table III.

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TABLE I. Electrical ratings.

Load	28 V dc sea level and 100,000 ft.		
	-01	-02	-03 through -08
	(Amps)	(Amps)	(Amps)
Resistive	5	10	7
Inductive	3	3	4
Motor	4	6	4

Ⓓ TABLE II. Qualification inspection-group submission.

Examination or test	Basic switch	Other switch samples	Extent of approval
Qualification inspection table of MIL-S-8805	M8805/80-01	M8805/80-02 M8805/80-04 2 samples each for each electrical test. M8805/80-02 M8805/80-03 M8805/80-04 M8805/80-06 M8805/80-07 1 each shall be submitted to following test sequence. 1. Strength of actuator means 2. Vibration 3. Shock 4. Operating characteristics	All

TABLE III. Part numbers.

Part number	Figure	"A" ±.030 (.76)	"B" max Ⓓ	Number of lead wires	Lead wire size	Circuitry type (see figure 4)
M8805/80-01	1	1.000 (25.4)	.980 (24.89)	6	20 AWG	A
-02	1	1.500 (38.1)	1.700 (43.18)	8	18 AWG	C
-03	1	.688 (17.48)	.980 (24.89)	6	20 AWG	A
-04	1	1.000 (25.4)	1.500 (30.48)	12	20 AWG	B
-05	2	.688 (17.48)	1.100 (26.67)	6	20 AWG	A
-06	2	1.000 (25.4)	1.250 (31.75)	12	20 AWG	B
-07	3	.688 (17.48)	1.100 (26.67)	6	20 AWG	A
-08	3	1.000 (25.4)	1.250 (31.75)	12	20 AWG	B

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CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activity:
Air Force - 11, 99

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5930-F647)

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